# ASFR-Net: Adversarial Alignment and Spatio-Frequency Refinement Network for Heterogeneous Remote Sensing Image Change Detection

Xin-Jie Wu, Zhi-Hui You, Si-Bao Chen, Qing-Ling Shu, Xiao Wang, Jin Tang, and Bin Luo

***Abstract*—The core challenge of heterogeneous change detection (CD) in remote sensing (RS) imagery lies in effectively decoupling genuine land-cover changes from significant modal disparities caused by distinct imaging mechanisms. These intrinsic inconsistencies are prone to introducing pseudochanges, thereby constraining detection accuracy. To address this, we propose a novel, end-to-end adversarial spatio-frequency refinement network (ASFR-Net). Initially, a modality-invariant representation learner (MIR-Learner) guides the backbone to extract modality-invariant features, effectively bridging the primary domain gap. Subsequently, to address persistent residual modal differences, we design an innovative spatio-frequency synergistic enhancement module (SFEM), which identifies and suppresses sensor-specific noise and artifacts that are difficult to discern in the spatial domain by leveraging frequency-domain processing. Multilevel difference features are then computed from these refined representations and fed into a decoder equipped with cascaded hierarchical guided fusion module (HGFM) blocks to generate precise change maps. To alleviate the data scarcity in heterogeneous tasks, we construct and release a new high-resolution benchmark specifically focused on building changes: the visible-near-infrared heterogeneous CD (VisNIR-HCD) dataset. It presents unique scientific challenges arising from deceptive visual similarity and nonlinear spectral inversions, providing a robust platform for evaluating model generalization. Extensive experiments on VisNIR-HCD and public datasets demonstrate that ASFR-Net achieves state-of-the-art (SOTA) performance, significantly outperforming existing methods. The source code and the VisNIR-HCD dataset are publicly available at https://github.com/LuoYang2024/ASFR-Net**

***Index Terms*—Change detection (CD), domain adaptation (DA), frequency domain analysis, heterogeneous image, multimodal, remote sensing (RS).**

## I. Introduction

CHANGE detection (CD) stands as a fundamental cornerstone in remote sensing (RS) image interpretation, dedicated to discerning significant land-cover alterations by analyzing bitemporal images of the same geographical area acquired at different timestamps [1]. By delivering timely and precise insights into Earth surface dynamics, CD plays an indispensable role across a spectrum of critical applications, ranging from postdisaster damage assessment [2] and urban expansion monitoring [3] to ecosystem sustainability analysis. In recent years, the explosive proliferation of earth observation programs has led to a surge in high-resolution, multisource RS data. This data deluge has necessitated a paradigm shift in CD techniques, propelling the field from traditional algebra-based comparisons toward advanced, data-driven deep learning paradigms capable of robust feature representation and high-level semantic abstraction.

Conventionally, most CD methods rely on the assumption that bitemporal data are acquired by homologous sensors. Under this assumption, unchanged regions exhibit consistent spectral and textural features, allowing algorithms to detect changes based on direct feature distances. However, in real-world scenarios, acquiring high-quality homogeneous image pairs is often hindered by adverse weather conditions, cloud cover, or sensor revisit cycles [4]. To ensure continuous all-weather monitoring, it has become a prevalent trend to utilize multisource data captured by different sensors, such as combining optical images with synthetic aperture radar (SAR) data, or visible spectrum images with near-infrared (NIR) data [5]. This gives rise to the task of heterogeneous CD.

However, compared with homogeneous CD, heterogeneous CD confronts a significantly more intricate and fundamental challenge: modal heterogeneity. Images from different modalities possess distinct imaging mechanisms, which manifest as profound nonlinear radiometric differences and disparate feature distributions even for the same ground objects [6]. This implies that the mapping between heterogeneous images is not merely complicated but highly nonlinear, often involving intricate phenomena, such as spectral inversions, that defy

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any simple linear correlation. Consequently, direct comparison strategies foundational to homogeneous CD are rendered ineffective, as they inherently lack the capacity to distinguish between genuine semantic changes and the spurious differences caused by modal disparity [7], prompting researchers to rethink the problem from alternative perspectives, such as image matching, to bypass direct intensity comparisons [8].

Existing heterogeneous CD approaches generally tackle this challenge through two main technical routes: image-to-image translation and feature-level alignment. Translation-based methods [9], [10] transform heterogeneous images into a pseudohomogeneous domain. However, the generation process introduces geometric distortions or artifacts, leading to error accumulation in subsequent detection. Feature-level alignment methods, such as AFENet [11] and HeteCD [12], utilize adversarial learning or metric constraints to project features into a shared latent space. Despite the significant progress achieved by these feature-level alignment methods, they typically encounter three intrinsic limitations.

First, they face a severe alignment-discriminability tradeoff. Aggressive alignment often inadvertently minimizes the distance between changed source features and target background features, thereby eroding the semantic discriminability required to accurately detect changes [13]. Second, most current approaches operate solely in the spatial domain, neglecting frequency-domain priors. However, residual modal differences, such as sensor-specific noise patterns or spectral shifts, are often deeply entangled with semantic content in the spatial domain but are readily separable in the frequency domain [14]. Neglecting frequency information restricts the capability of the model to filter out stubbornly persistent modal noise that spatial convolutions struggle to resolve. Third, existing frameworks frequently overlook the semantic-spatial gap and the amplification of cross-modal noise during decoding. While recent homogeneous CD methods employ advanced multilevel aggregation strategies to recover details, applying these directly to heterogeneous tasks often fails. Shallow-level difference features in heterogeneous pairs are highly contaminated by modality-specific noise. Without dedicated selective filtering, existing aggregation modules indiscriminately propagate or even amplify this cross-modal noise into the final prediction. Leveraging aligned deep semantic priors to gate shallow details thus remains a bottleneck for precise boundary delineation. Furthermore, the development of supervised heterogeneous CD methods is constrained by the lack of extensive, high-resolution heterogeneous benchmarks, particularly for visible-NIR scenarios, which are crucial for urban vegetation and building monitoring. This data scarcity fundamentally restricts the training of robust deep models, often resulting in poor generalization when facing complex real-world spectral variations and diverse environmental conditions.

To address these challenges, we propose a novel end-to-end framework named the adversarial spatio-frequency refinement network (ASFR-Net). First, we design a modality-invariant representation learner (MIR-Learner) incorporating a gated adversarial domain unifier (GADU) and a polarity-aware feature regularizer (PAFR). GADU utilizes prediction maps to condition the alignment, ensuring semantic consistency, while the PAFR imposes geometric constraints to prevent feature space collapse, preserving discriminability. Second, to address residual modal discrepancies, we introduce a spatio-frequency synergistic enhancement module (SFEM). By transforming features into the Fourier domain, the SFEM identifies and suppresses high-frequency modal noise and low-frequency style biases imperceptible in the spatial domain, injecting purified frequency-domain priors back into the network. Finally, to overcome the semantic-spatial gap, a decoder equipped with cascaded hierarchical guided fusion module (HGFM) blocks is employed. HGFM uses a residual dynamic gating mechanism to leverage deep semantic priors to filter cross-modal noise from shallow difference features, preserving critical high-frequency details for precise change maps. Furthermore, we construct and release visible-NIR heterogeneous CD (VisNIR-HCD), a high-resolution benchmark for heterogeneous building CD.

The main contributions of this work are as follows.

1) We propose a robust heterogeneous CD method named ASFR-Net. By organically integrating global adversarial alignment with fine-grained spatio-frequency refinement, it explicitly decouples modality-induced discrepancies from genuine semantic changes. Comprehensive evaluations across diverse public benchmarks demonstrate that ASFR-Net yields state-of-the-art (SOTA) performance, decisively outperforming current competitive methods.
2) To bridge the primary domain gap, we design the MIR-Learner. It reconciles the intrinsic conflict between feature alignment and semantic discriminability via polarity-aware geometric constraints during adversarial training, effectively preventing feature space collapse while ensuring robust cross-modal semantic consistency.
3) To mitigate residual modal discrepancies, we introduce the SFEM. This module leverages frequency-domain priors to explicitly decouple and suppress sensor-specific high-frequency noise and low-frequency style biases, while preserving structural integrity and enhancing feature discriminability through dense aggregation.
4) To bridge the semantic-spatial gap during the decoding phase, we propose the HGFM. It employs a residual dynamic gating mechanism that utilizes deep semantic priors to actively filter cross-modal noise from shallow features for the precise delineation of change boundaries.
5) We construct and open-source a high-resolution visible-NIR heterogeneous CD dataset named VisNIR-HCD. Focused on building changes, it presents unique scientific challenges arising from deceptive visual similarity and nonlinear spectral inversions, thereby providing a rigorous platform for evaluating model generalization.

## II. Related Work

### A. Heterogeneous CD

Traditional CD research typically explores surface dynamics by assuming data homogeneity between bitemporal image pairs. Under this premise, homogeneous CD generally dominates the field, as unchanged regions exhibit consistent

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spectral and textural features, allowing for direct feature comparison based on distance metrics. However, in practical applications such as disaster assessment [2], postearthquake damage evaluation, and continuous all-weather monitoring [4], environmental constraints such as severe meteorological disturbances, heavy cloud occlusions, and prolonged satellite revisit intervals inherently restrict the collection of ideal homogeneous image pairs. While visible light spectral images capture rich textural and boundary details, they suffer from severe information degradation under poor visibility. Conversely, SAR provides all-weather capabilities but introduce severe geometric distortions and speckle noise that obscure building structures. Meanwhile, NIR sensors effectively mitigate atmospheric haze and strongly suppress seasonal vegetation interference, yet still exhibit massive spectral disparities compared to visible bands. To ensure continuous all-weather Earth observation, utilizing multisource data captured by different sensors has become an inevitable trend [5]. This transition fundamentally gives rise to the highly challenging task of heterogeneous CD.

Heterogeneous CD confronts a profound obstacle: modal heterogeneity. Distinct imaging mechanisms manifest as severe nonlinear radiometric differences, structural inconsistencies, and disparate feature distributions, even for the identical geographical objects [6]. Consequently, direct comparison strategies are rendered utterly ineffective, as they inherently lack the capacity to decouple genuine semantic changes from spurious pseudochanges caused by modal disparity [15]. To address this, early research predominantly explored image-to-image translation strategies. These methods aim to map the radiometric characteristics of one modality into another via image regression or generative adversarial networks (GANs), thereby creating pseudohomogeneous pairs [9], [10], [16]. While conceptually intuitive, the generation process itself frequently introduces geometric distortions, semantic shifts, or synthetic artifacts. These flaws inevitably propagate to downstream detection tasks and severely undermine the reliability of the final results [17]. This limitation has spurred a significant paradigm shift toward direct feature-level alignment.

### B. Adversarial Learning for Feature Alignment

To circumvent the aforementioned pitfalls of image translation, feature-level alignment methods attempt to project high-level representations from disparate domains into a shared, modality-invariant latent space to achieve semantic consistency. Among these, adversarial learning-based domain adaptation (DA) approaches have recently gained immense prominence due to their adaptive capacity in modeling highly complex and nonlinear multimodal relationships [11], [13].

By establishing a min–max game between a feature extractor and a domain discriminator, these methods compel the network to produce modality-invariant representations. For instance, AFENet [11] proposes an adversarial feature equilibrium network that utilizes multiscale adversarial DA to reduce the modal gap. It incorporates a divergence-aware contrastive module (DCM) as a regularizer to dynamically adjust feature differences, thereby maintaining representation consistency while maximizing the discriminability of change targets. Similarly, CMMAN [13] introduces a change-masked modality alignment network through a multitask framework. It elegantly leverages the change probability map predicted by the CD branch to mask change regions during the image modal transformation (IMT) process, thereby explicitly decoupling object changes from modal heterogeneity. Despite the powerful alignment capabilities demonstrated by these models, adversarial learning frequently confronts an intrinsic alignment-discriminability tradeoff. Aggressive global distribution alignment can inadvertently force changed source regions to align with target background features, thereby eroding the semantic distinctiveness required to accurately identify subtle changes [18]. Reconciling this conflict requires complementary perspectives beyond spatial-domain adversarial alignment.

### C. Frequency-Domain Analysis

Most existing heterogeneous CD frameworks operate solely in the spatial domain, overlooking the decoupling capabilities of frequency-domain priors. Recently, frequency-domain analysis has demonstrated substantial potential in the field of RS, proving effective at disentangling image characteristics, with style biases in low frequencies and structural details in high frequencies [14]. Initially, models leaned toward leveraging high-frequency priors to enhance structural integrity. For instance, HFA-Net [19] incorporates a built-in high-frequency attention block (HFAB) that explicitly extracts and highlights high-frequency information via isotropic Sobel operators and spatialwise attention, improving the detection of sharp building boundaries. To tackle pseudochanges caused by varying imaging conditions, FIMP [14] employs a Fourier feature interaction strategy. By transforming bitemporal features into the frequency domain and applying an adaptive frequency filtering module (AFFM), this method selectively weights different frequency components to mitigate the impact of style discrepancies. Focusing on adaptive spectrum decoupling in the latent space, FeaSpect [20] utilizes extraction boxes with adaptive attention and learnable strides to align amplitude spectra, effectively eliminating temporal style shifts while meticulously preserving discriminative semantics. More recently, WaveHFG [21] leverages wavelet transforms to extract shared high-frequency structural information, guiding the network to learn modality-invariant representations.

Despite these exciting advancements, the application of frequency-domain analysis to explicitly resolve modal heterogeneity in multisource CD remains largely underexplored. Residual modal discrepancies, which are deeply entangled with semantic content in the spatial domain, are often readily separable in the Fourier domain. Inspired by this insight, our ASFR-Net leverages the SFEM alongside global adversarial alignment to explicitly filter out stubborn sensor-specific artifacts, thereby ensuring optimal feature discriminability.

## III. Proposed Benchmark Dataset

In this study, we construct a new benchmark dataset termed VisNIR-HCD to address the lack of high-resolution heterogeneous data. As illustrated in Fig. 1, the dataset

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Fig. 1. Annotation samples from the proposed VisNIR-HCD dataset. The right side shows a close-up of the region with changes in the sample and highlights the challenge of distinguishing buildings due to nonlinear spectral differences.

TABLE I
SUMMARY OF WIDELY USED CD DATASETS AND THE PROPOSED VISNIR-HCD

| Dataset | Resolution (m/pixel) | Image pairs | Image size | Train/Val/Test | $T_1/T_2$ type | Change Ratio | Change Instances |
|---|---|---|---|---|---|---|---|
| Homogeneous change detection datasets | | | | | | | |
| LEVIR-CD [3] | 0.5 / 0.5 | 637 | $1024 \times 1024$ | 445/64/128 | RGB/RGB | 4.63% | 31,333 |
| WHU-CD [22] | 0.2 / 0.2 | 1 | $32207 \times 15354$ | 5947/743/744 | RGB/RGB | 4.33% | 12,796 |
| Heterogeneous change detection datasets | | | | | | | |
| MT-Wuhan [18] | 10 / 3 | 1 | $11216 \times 13693$ | 552/129/112 | RGB/SAR | 15.31% | 4,204 |
| XiongAn [12] | 4 / 8 | 2314 | $512 \times 512$ | 1901/413/413* | RGB/SAR | 2.00% | 16,846 |
| Proposed heterogeneous change detection dataset | | | | | | | |
| **VisNIR-HCD (Ours)** | 0.8 / 0.8 | **8,432** | $256 \times 256$ | 5901/839/1692 | **RGB/NIR** | **3.08%** | **18,488** |

*Note: Due to discrepancies between the paper description and the released dataset, statistics follow the actual public data where the validation set serves as the test set.

encompasses diverse scenes exhibiting significant spectral differences between the visible and NIR modalities. The data collection covers two representative regions in Wuhan, China, spanning 114.04°E–114.23°E, 30.36°N–30.61°N and 114.32°E–114.53°E, 30.50°N–30.72°N. Images were acquired by the PMS1 and PMS2 sensors on the Gaofen-2 satellite, featuring diverse landscapes such as urban residential zones, industrial lands, and rural settlements. All images are aligned to the WGS 84/UTM zone 50-N coordinate system, achieving a spatial resolution of 0.8 m/pixel. The dataset uses bitemporal image pairs that align precisely in space. Each pair consists of a prechange visible red-green-blue (RGB) image from 2016 for the prechange phase $T_1$ and a single-band NIR image from 2023 for the postchange phase $T_2$. To provide a comprehensive statistical profile, the VisNIR-HCD dataset comprises about 552.6 million pixels and a total of 18 488 building change instances. The foreground changed regions account for only 3.08% of the total, while the unchanged background dominates at 96.92%. This high foreground-to-background ratio reflects the inherent physical sparsity of buildings in geographic space and aligns closely with the distributions observed in other mainstream benchmarks, such as XiongAn [12] at 2.00% and WHU-CD [22] at 4.33%.

Table I provides a comprehensive comparison between the proposed VisNIR-HCD and existing mainstream CD datasets. Current benchmarks can be broadly categorized into homogeneous and heterogeneous types. Homogeneous datasets, such as LEVIR-CD and WHU-CD, rely exclusively on RGB imagery. While they offer high spatial resolution, they lack the spectral diversity required to evaluate cross-modal algorithms. Conversely, existing heterogeneous datasets like MT-Wuhan and XiongAn predominantly focus on the optical-SAR modality. While SAR offers critical all-weather observation capabilities, it introduces severe speckle noise, complex multiple scattering effects, and geometric distortions like layover and foreshortening. These factors inherently obscure structural edges and degrade building extraction performance. On the contrary, the proposed VisNIR-HCD dataset focuses on the under-explored visible-NIR modality, offering distinct practical benefits. NIR imagery maintains high geometric fidelity and spatial resolution comparable to visible light while providing enhanced atmospheric penetration to mitigate thin clouds or haze. Most importantly, NIR effectively suppresses seasonal vegetation interference, which is a primary source of pseudochanges in building CD. Because healthy vegetation strongly reflects NIR while man-made structures absorb it, RGB-NIR combinations serve as a highly reliable tool for decoupling seasonal variations from genuine structural changes.

RGB-NIR presents a unique challenge distinct from optical-SAR tasks, namely, deceptive similarity. Because RGB and NIR images share high structural consistency and visual clarity, models are prone to erroneously assuming a simple linear mapping between them. However, the true mapping involves

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complex and category-specific nonlinear spectral inversions. Methods tailored for optical-SAR often employ aggressive global distribution alignment to bridge massive modality gaps. When applied directly to RGB-NIR data, such approaches can trigger overalignment or negative transfer. This inadvertently erases the fine-grained spectral cues essential for distinguishing pseudochanges from genuine building alterations.

To ensure both spectral consistency and annotation accuracy, a rigorous multistage processing pipeline was implemented. First, the original images underwent radiometric correction and normalization to standardize pixel intensities to the range of $[0, 255]$. Following this preprocessing, a meticulous annotation process was conducted on the large-format and coregistered image pairs. To mitigate visual ambiguities arising from modal heterogeneity, an auxiliary optical reference strategy was employed. By leveraging a corresponding pair of contemporaneous high-resolution optical images as a reference, precise building change labels were generated. These initial labels were then subjected to a dual-person cross-validation protocol, serving as a critical quality control step to filter low-quality samples and ensure high fidelity. Subsequently, these fully annotated and validated large-scale images were partitioned into nonoverlapping patches of $256 \times 256$ pixels. The final dataset comprises 8432 high-quality sample pairs and is randomly partitioned into a training set with 5901 pairs, a validation set with 839 pairs, and a test set with 1692 pairs. This strictly follows a 7:1:2 ratio to facilitate a rigorous evaluation of model generalization performance.

Furthermore, to facilitate broader research in homogeneous and heterogeneous CD, as well as multimodal fusion, we simultaneously open source the complete quad-image sets containing RGB and NIR modalities at both $T_1$ and $T_2$.

## IV. Proposed Approach

In this section, the overall architecture of the proposed ASFR-Net is first briefly described, and then the intrinsic mechanisms of its constituent modules are elaborated in detail. Finally, the optimization strategy is presented.

### A. Approach Overview

The schematic architecture of ASFR-Net is illustrated in Fig. 2. Built upon a cascaded feature refinement paradigm, the network synergizes adversarial feature alignment with spatio-frequency enhancement to tackle heterogeneous CD tasks through three key modules: a Siamese encoder guided by a MIR-Learner, a SFEM, and a decoder equipped with cascaded HGFM blocks.

*1) Encoder and Adversarial Alignment:* A weight-sharing Siamese MobileNetV2 [23] is employed to extract multilevel hierarchical features from heterogeneous images $I_{T_1}$ and $I_{T_2}$. The extracted feature sets are denoted as $\mathcal{F}_t = \{F_t^1, \ldots, F_t^5\}$, where $t \in \{T_1, T_2\}$. We avoid independent encoders to prevent overfitting to sensor-specific noise. Instead, sharing weights acts as a strict physical regularizer, mapping disparate inputs into a roughly overlapping latent space from the outset. This reduces the burden on the subsequent adversarial learning stage, as the discriminator only needs to refine a unified feature distribution, thereby stabilizing gradient updates. To bridge the modality gap, the MIR-Learner integrates an adversarial mechanism [24] to constrain the encoder to generate modality-invariant representations. However, since aggressive alignment may inadvertently suppress high-frequency structural details critical for delineating boundaries, a subsequent feature refinement stage is necessitated.

*2) Spatio-Frequency Synergistic Enhancement:* Structural integrity resides in high-frequency components, which are often attenuated during adversarial training [13]. The SFEM mitigates the loss of feature discriminability and detail suppression caused by adversarial alignment. It integrates an adaptive spectral attention unit (ASAU) to suppress modality-specific noise, and a holistic pyramid aggregation unit (HPAU) for spatial aggregation. By injecting the purified frequency priors from ASAU into HPAU, SFEM restores fine-grained structural details, yielding the refined feature maps $S_t^i$

$$S_t^i = \mathrm{SFEM}_i\left(\mathcal{F}_{T_1}, \mathcal{F}_{T_2}\right), t \in \{T_1, T_2\}, i \in \{1, \ldots, 5\} \quad (1)$$

where $S_t^i$ denotes the structurally enhanced feature at the $i$th level for the image at time $t$. To generate the change representation, we compute the multilevel difference features

$$X^i = \mathrm{Abs}\left(S_{T_1}^i - S_{T_2}^i\right), \quad i \in \{1, \ldots, 5\} \quad (2)$$

where $\mathrm{Abs}(\cdot)$ denotes the elementwise absolute difference operation between the bitemporal features, and $X^i$ represents the difference feature map at the $i$th level.

*3) Decoder:* The decoder comprises cascaded HGFM blocks designed to progressively bridge the gap between global context and local details. To ensure precise object localization and robust suppression of pseudochanges, each HGFM leverages deep semantic priors to adaptively guide the refinement of shallower, detail-oriented difference features $X^i$

$$P_i, M_i = \mathrm{HGFM}\left(X^i, \mathrm{Up}\left(P_{i+1}\right)\right), \quad i \in \{4, \ldots, 1\} \quad (3)$$

where $\mathrm{Up}(\cdot)$ denotes the bilinear upsampling operation. $P_i$ represents the refined semantic feature, and $M_i$ denotes the corresponding change probability map generated at the $i$th level. Thresholding the highest-resolution probability map $M_1$ at 0.5 yields the final binary change map. During training, deep supervision constrains predictions across all scales.

### B. Modality-Invariant Representation Learner

In heterogeneous CD, decoupling genuine semantic changes from modal disparities is a fundamental challenge. To address this, MIR-Learner, illustrated in Fig. 3, employs an adversarial strategy, ensuring learned features remain invariant to sensor distributions while capturing intrinsic structural changes. This is achieved via two synergistic components: GADU for distribution alignment and PAFR for preserving discriminability. During training, multiscale features from the weight-sharing encoder are simultaneously fed into both modules. Operating in parallel, they form a complementary push-pull optimization strategy. The GADU aggressively aligns global feature distributions to eliminate the macroscopic modality gap, while PAFR operates in parallel to impose explicit geometric boundary constraints, preventing overalignment and strictly

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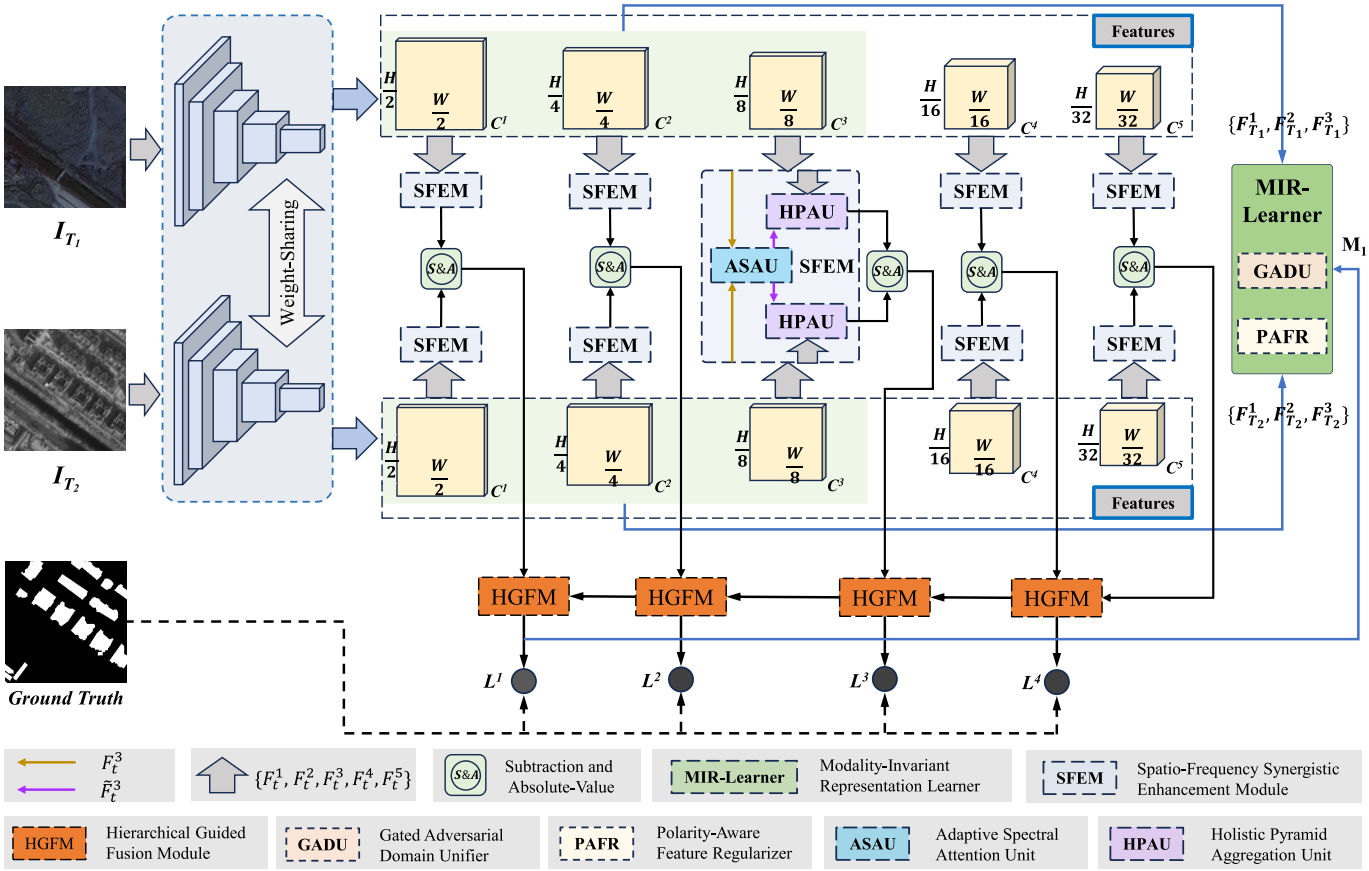



Fig. 2. Overview of the proposed ASFR-Net. First, a weight-sharing Siamese encoder extracts hierarchical bi-temporal features from the heterogeneous image pair. Then, the MIR-Learner, incorporating the GADU and the PAFR, is utilized to bridge the domain gap and align cross-modal features. Next, the SFEM, whose detailed structure is exemplified at the third level, leverages the ASAU and the HPAU to suppress residual modal noise and refine feature discriminability. Finally, the multiscale guided fusion decoder employs cascaded HGFM blocks to progressively aggregate multilevel information for precise change map generation. $(H, W)$ denotes the height and width of original input, and $C^i$ explicitly indicates the channel dimension at the $i$th level.

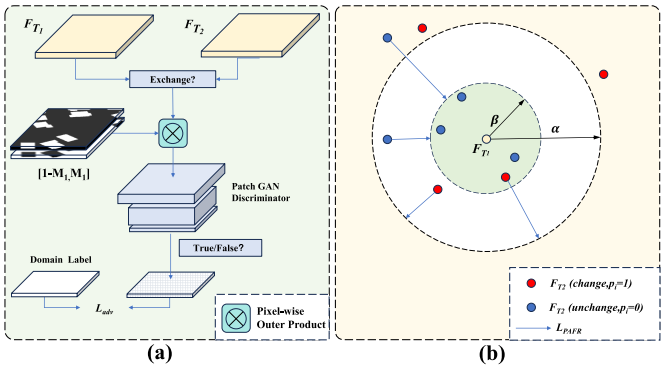



Fig. 3. Overview of the MIR-Learner. (a) GADU: utilizes $M_1$ predicted by the decoder to condition feature alignment via adversarial learning. The domain label acts as a binary indicator specifying whether the input features originate from the source or target domain. (b) PAFR: regularizes feature space geometry by constraining unchanged features within distance $\beta$ and pushing changed features beyond distance $\alpha$.

preserving local semantic discriminability. The gradients from their respective objectives are then jointly backpropagated to optimize the shared encoder.

*1) Gated Adversarial Domain Unifier:* The primary objective of GADU is to construct a modality-invariant unified feature space. To address the risk of negative transfer caused by erroneously forcing changed source regions to align with target background regions [11], GADU incorporates a prediction-based conditional adversarial mechanism to enforce semantically consistent alignment.

In particular, we utilize the change probability map $M_1$ predicted by the CD branch to condition feature alignment across the multiscale feature representations $\{F^i_{T_1}, F^i_{T_2}\}^3_{i=1}$. To capture category-conditional information, $M_1$ is expanded into a two-channel probability map $[1 - M_1, M_1]$, representing the predicted probabilities of unchanged and changed classes, respectively. Without loss of generality, we treat the $T_1$ branch as the source domain $\mathcal{S}$ and the $T_2$ branch as the target domain $\mathcal{T}$. For the $i$th feature level, features from the source domain $F^i_{\mathcal{S}}$ and target domain $F^i_{\mathcal{T}}$ are conditioned via a pixelwise outer product with the spatially aligned probability map

$$H^i_d = F^i_d \otimes \mathrm{AMP}_i\left([1 - M_1, M_1]\right), \quad d \in \{\mathcal{S}, \mathcal{T}\} \tag{4}$$

where $H^i_d$ denotes the conditioned feature map for domain $d$, and $\mathrm{AMP}_i(\cdot)$ denotes the adaptive max pooling operation employed to spatially align the dimensions of $[1 - M_1, M_1]$ to match the spatial resolution of the $i$th feature level. This explicitly conditions the multimodal feature distribution on

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specific semantic categories, preserving semantic consistency throughout the alignment process.

To optimize alignment, a multiscale domain discriminator $D = \{D_i\}_{i=1}^{3}$ evaluates the conditioned joint features at each corresponding scale. To ensure training stability and gradient smoothness, $D$ is optimized via a Hinge adversarial loss augmented with an $R_1$ gradient penalty. Unlike standard binary cross-entropy (BCE)-based objectives used in AFENet [11], this formulation provides stable gradient dynamics and mitigates mode collapse; the specific discriminator optimization objective $\mathcal{L}_D$ is detailed in Section IV-E. Conversely, for the generator, the scale-specific adversarial objective $\mathcal{L}_{\text{adv}}^{i}$ drives the target domain features $H_{\mathcal{T}}^{i}$ to deceive the scale-specific discriminator $D_i$ by minimizing the likelihood of correct domain identification

$$\mathcal{L}_{\text{adv}}^{i} = -\mathbb{E}\left[D_i\left(H_{\mathcal{T}}^{i}\right)\right], \quad i \in \{1,2,3\}. \tag{5}$$

By minimizing $\mathcal{L}_{\text{adv}}^{i}$, the generator is compelled to extract modality-invariant features that deceive $D_i$, thereby bridging the domain gap between the source and target domains.

*2) Polarity-Aware Feature Regularizer:* Despite the effectiveness of GADU in domain alignment, it risks overcompressing the feature manifold, potentially compromising semantic discriminability. This tradeoff is addressed by PAFR, which enforces geometric regularity and enhances separability in the latent space. Distinct from the DCM in AFENet [11], which primarily targets divergence amplification, PAFR functions as a geometric manifold regularizer designed to explicitly preserve the topological integrity of the feature space.

Formally, for the $i$th feature level, PAFR imposes polarity-dependent constraints on the Euclidean distance between $L_2$-normalized bitemporal features $F_{\mathcal{S}}^{i}$ and $F_{\mathcal{T}}^{i}$. Guided by change labels, we enforce distinct margins: feature pairs in changed regions ($p_n = 1$) must separate beyond a lower bound $\alpha$ (repulsion), whereas those in unchanged regions ($p_n = 0$) are constrained within an upper bound $\beta$ (attraction). Consequently, the PAFR objective for the $i$th scale level is formulated as a bounded, ReLU-based margin loss

$$\mathcal{L}_{\text{PAFR}}^{i} = \frac{1}{N_i}\sum_{n=1}^{N_i}\left[p_n\phi\left(\alpha - \delta_n^{i}\right) + (1-p_n)\,\phi\left(\delta_n^{i} - \beta\right)\right] \tag{6}$$

where $\delta_n^{i} = \|F_{\mathcal{S}}^{i}(n) - F_{\mathcal{T}}^{i}(n)\|_2$ denotes the pixelwise Euclidean distance at spatial location $n$, $N_i$ is the total number of spatial pixels at the $i$th level, and $\phi(\cdot)$ represents the ReLU$(\cdot)$ activation function. Since the learned features are $L_2$-normalized, setting an appropriate repulsion margin $\alpha$ geometrically guarantees a sufficient angular separation between changed feature pairs, establishing a robust and clear decision boundary in the latent space. Meanwhile, introducing a slightly relaxed attraction margin $\beta$, rather than a strict zero tolerance, enforces the tight clustering of unchanged features while explicitly preventing the network from overpenalizing inherent modality noise. The exact numerical configurations of these boundaries and their empirical validation are deferred to the hyperparameter sensitivity analysis in Section V-E.

To strike an optimal balance between modality alignment and semantic discriminability, a synergistic optimization strategy is adopted. Instead of treating these modules in isolation, it integrates the multiscale adversarial loss from GADU and the geometric contrastive loss from PAFR as joint auxiliary regularizers. These terms operate alongside the primary pixel-level CD objective to constitute the total generator loss. The detailed mathematical formulation is provided in Section IV-E.

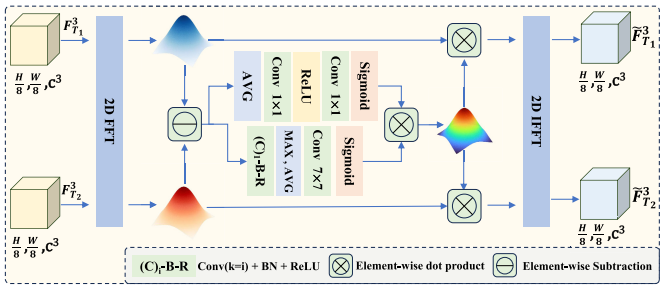



Fig. 4. Overview of the ASAU. It leverages FFT to transform features into the frequency domain, employs a difference-guided attention mask to filter modal noise, and reconstructs the refined features via IFFT.

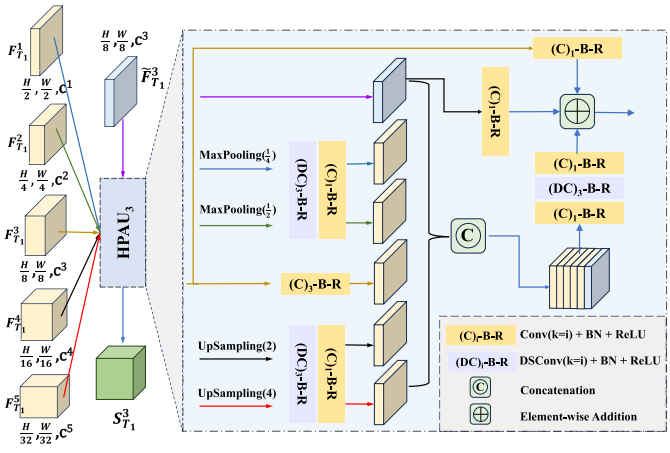



Fig. 5. Overview of the HPAU. Leveraging the frequency-refined features from ASAU as guidance, it densely aggregates hierarchical spatial features to reconstruct modality-invariant representations with rich structural details.

## C. Spatio-Frequency Synergistic Enhancement Module

While adversarial alignment mitigates global shifts, residual low-frequency biases and high-frequency artifacts persist. To address this, the SFEM employs frequency priors to guide spatial reconstruction, moving beyond spatial-domain-only processing. In our context, modal noise is operationally defined as task-irrelevant, non semantic spectral components that are strictly induced by disparate sensor mechanisms. This includes high-frequency sensor artifacts, such as SAR speckle or periodic stripes, and low-frequency global biases, such as atmospheric variations or illumination shifts. These components typically lack coherent semantic structure and are precisely what our frequency-domain attention is designed to target. The SFEM first filters spectral artifacts from bitemporal pyramids ($\mathcal{F}_{T_1}, \mathcal{F}_{T_2}$) via ASAU (Fig. 4), then performs spectral-guided spatial aggregation with HPAU (Fig. 5). Here, we exemplify the detailed internal structures of the ASAU and the HPAU using the third-level feature processing pipeline of the $T_1$ branch.

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*1) Adaptive Spectral Attention Unit:* Pseudochanges driven by seasonality or illumination exhibit distinct spectral patterns: style biases dominate low frequencies, while noise affects high frequencies. Leveraging this, ASAU adaptively suppresses these task-irrelevant components [20].

Currently, applying off-the-shelf frequency-domain modules directly to heterogeneous tasks often yields suboptimal results, as they struggle to bridge the inherent nonlinear radiometric inversions and structural inconsistencies. To overcome this, we leverage the Fourier transform to decouple style and content, decomposing the previously defined modal noise into specific frequency bands, thus making frequency-domain differencing reliable for heterogeneous CD. To ensure the ASAU addresses modality-induced discrepancies without removing true change signals, we introduce a dynamic, difference-guided dual attention mechanism. Unlike traditional filters relying on manually fixed cutoffs, our frequency attention gates are adaptively computed based on the spectral difference map. This input-dependent design allows the parallel channel and spatial attention branches to dynamically recalibrate each frequency component. Genuine land-cover changes produce robust, spatially coherent difference signals, which the network adaptively preserves while suppressing modal artifacts. This adaptive decoupling is difficult to achieve in the spatial domain, where style and content are intrinsically entangled, thus validating the reliability of our frequency-based approach.

Given spatial features $F^i_{T_1}, F^i_{T_2} \in \mathbb{R}^{H^i \times W^i \times C^i}$, we first perform a 2-D fast Fourier transform (FFT) to compute the spectral difference map $D^i_{\text{freq}}$

$$D^i_{\text{freq}} = \text{FFT}\left(F^i_{T_1}\right) - \text{FFT}\left(F^i_{T_2}\right). \tag{7}$$

To disentangle artifacts, a parallel dual-spectral attention mechanism processes the concatenated real and imaginary parts of $D^i_{\text{freq}}$, denoted as $D^i_{\text{cat}}$. In particular, the frequency channel attention (FCA) branch models interchannel dependencies by passing the globally average-pooled (AVG) descriptor through a two-layer multilayer perceptron (MLP) to generate the channel gate $G^i_c$

$$G^i_c = \sigma\left(\text{MLP}\left(\text{AVG}\left(D^i_{\text{cat}}\right)\right)\right) \tag{8}$$

where $\sigma(\cdot)$ denotes the sigmoid function. Simultaneously, the frequency spatial attention (FSA) branch localizes noise by applying a large-kernel convolution on channel-pooled features to generate the spatial gate $G^i_s$

$$G^i_s = \sigma\left(\text{Conv}_{7\times7}\left(\text{Cat}\left[\text{MaxP}_c\left(\text{Conv}_{1\times1}\left(D^i_{\text{cat}}\right)\right)\right.\right.\right. \\ \left.\left.\left.\text{AvgP}_c\left(\text{Conv}_{1\times1}\left(D^i_{\text{cat}}\right)\right)\right]\right)\right) \tag{9}$$

where $\text{MaxP}_c$ ($\text{AvgP}_c$) denotes channelwise max (average) pooling, and the batch normalization and ReLU activation functions following the $1 \times 1$ convolution are omitted for brevity. The unified gate $G^i_{\text{total}}$ synergistically combines the channel and spatial gates to adaptively recalibrate the features

$$G^i_{\text{total}} = G^i_c \odot G^i_s \tag{10}$$

where $\odot$ denotes elementwise multiplication. This unified gate adaptively filters the original spectra to yield purified priors via inverse FFT

$$\tilde{F}^i_t = \text{IFFT}\left(\text{FFT}\left(F^i_t\right) \odot G^i_{\text{total}}\right), \quad t \in \{T_1, T_2\}. \tag{11}$$

This rectifies modal discrepancies in the frequency domain, yielding high-fidelity structural priors for spatial enhancement.

*2) Holistic Pyramid Aggregation Unit:* Following spectral purification, spatial features often suffer from structural degradation due to aggressive domain alignment. To address this, we propose the HPAU. Unlike standard spatial-only pyramids, HPAU introduces a frequency-guided holistic view that leverages purified priors from ASAU to mitigate modal disparities and restore multiscale spatial details.

For $j$th level at time $T_1$, HPAU aggregates the feature hierarchy $\mathcal{F}_{T_1}$ guided by the frequency prior $\tilde{F}^j_{T_1}$. First, cross-scale alignment unifies the resolutions of auxiliary features $\{F^k_{T_1}\}_{k\neq j}$ to $j$th level, yielding aligned contexts $\{F^{j\leftarrow k}_{T_1}\}$. To reconstruct modality-invariant structures, we employ a dual-injection strategy. In the first injection, purified frequency priors are concatenated with spatial contexts to suppress modal noise during dense aggregation. Subsequently, to preserve high-frequency details blurred by heterogeneity, the second injection employs a residual mechanism. This successive refinement process is formulated as

$$S^j_{T_1} = \text{Conv}_{1\times1}\left(\text{DSConv}_{3\times3}\left(\text{Conv}_{1\times1}\left(\text{Cat}\right.\right.\right. \\ \left.\left.\left.\left[\left\{F^{j\leftarrow k}_{T_1}\right\}^5_{k=1,k\neq j}, F^j_{T_1}, \tilde{F}^j_{T_1}\right]\right)\right)\right) \\ + \text{Conv}_{1\times1}\left(F^j_{T_1}\right) + \text{Conv}_{1\times1}\left(\tilde{F}^j_{T_1}\right) \tag{12}$$

where Cat[·] and $\text{DSConv}_{3\times3}(\cdot)$ denote concatenation operation and depthwise separable convolution, respectively. The batch normalization and ReLU activation function are omitted.

By synergizing multiscale semantics with high-fidelity structural priors, HPAU yields a robust, modality-agnostic representation $S^j_{T_1}$ for subsequent processing. Crucially, the residual connection that reinjects the original, unfiltered spatial features serves as a second critical safeguard, preventing the loss of fundamental structures and fine-grained boundary details during frequency filtering.

### D. Hierarchical Guided Fusion Module

A central bottleneck in the decoding phase of heterogeneous CD is bridging the semantic–spatial gap without propagating cross-modal noise. While deep features provide robust semantics, shallow features contain fine details but suffer from severe modal disparities. Standard decoders often passively fuse these, inadvertently propagating noise. To reconcile this, HGFM leverages deep semantic priors to actively filter and refine shallow difference features, suppressing modality-induced pseudochanges. Initially, the deepest difference feature $X^5$ generates the coarse-grained semantic prior feature $P_5$

$$P_5 = \text{Conv}_{3\times3}\left(X^5\right) \tag{13}$$

where batch normalization and the ReLU activation function are omitted. Subsequently, a top-down iterative fusion executes from level $i = 4$ down to 1. At each stage, HGFM refines the difference feature $X^i$ using guidance from the upsampled higher level feature $P_{i+1}$, yielding the refined semantic feature $P_i$ and the corresponding change probability map $M_i$.

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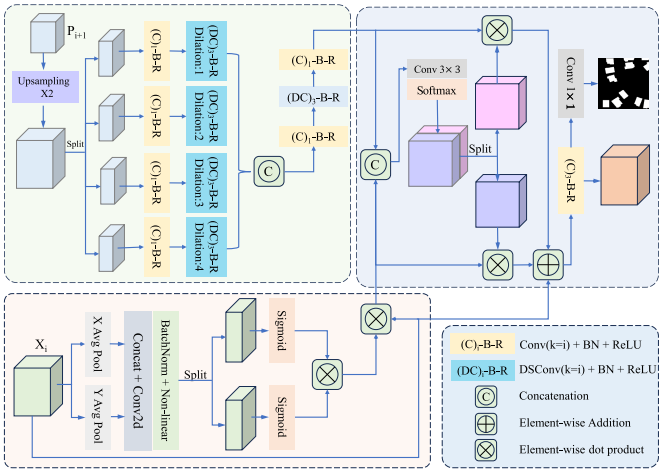



Fig. 6. Structure of the HGFM. This module uses a residual dynamic gating mechanism to adaptively fuse multiscale semantic priors from dilated convolutions with spatial details from coordinate attention.

HGFM, illustrated in Fig. 6, serves as the core execution unit designed to seamlessly integrate the shallow detail feature $X^i$ and the deep guidance feature $\mathrm{Up}(P_{i+1})$. Unlike simple concatenation, the module operates through a coherent enhance-then-fuse strategy involving two parallel streams.

On the semantic side, to harvest robust multiscale contextual priors from the guidance feature, we employ a semantic split-aggregation strategy inspired by RFANet [25]. The input channels are partitioned into multiple groups, each processed by a parallel atrous convolution branch with a distinct dilation rate. This design enables the capture of diverse receptive fields. To efficiently consolidate these features, a depthwise separable fusion block is applied to generate the unified context feature

$$F_{\mathrm{ctx}} = \mathrm{DSConv}_{3\times3}\left(\mathrm{Cat}\left[o_1, \ldots, o_{N_g}\right]\right) \tag{14}$$

where $o_g$ denotes the multiscale feature from the $g$th parallel atrous convolution branch and $\mathrm{Cat}[\cdot]$ represents the concatenation operation. $\mathrm{DSConv}_{3\times3}(\cdot)$ indicates the depthwise separable convolution used to fuse the concatenated sub-features; batch normalization and activation functions are omitted.

In parallel, since the shallow feature $X^i$ is detail-rich but noise-prone, coordinate attention [26] aggregates it horizontally and vertically into direction-aware maps $(A_h, A_w)$ to highlight change-relevant structures

$$\tilde{X}^i = X^i \odot A_h \odot A_w. \tag{15}$$

Next, a dynamic gating mechanism learns adaptive weights $[W_{\mathrm{det}}, W_{\mathrm{ctx}}]$ from their concatenation to optimally fuse the semantic context $F_{\mathrm{ctx}}$ and enhanced details $\tilde{X}^i$. The fused feature adds a residual connection to the original $X^i$ and passes through a convolutional layer to yield the refined semantic feature $P_i$, which preserves high-frequency boundaries and guides the subsequent decoding stage

$$[W_{\mathrm{det}}, W_{\mathrm{ctx}}] = \mathrm{Softmax}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Cat}\left[\tilde{X}^i, F_{\mathrm{ctx}}\right]\right)\right) \tag{16}$$

$$P_i = \mathrm{Conv}_{3\times3}\left(\left(W_{\mathrm{det}} \odot \tilde{X}^i\right) + \left(W_{\mathrm{ctx}} \odot F_{\mathrm{ctx}}\right) + X^i\right) \tag{17}$$

where $\mathrm{Cat}[\cdot]$ indicates the concatenation operation and batch normalization and activation functions are omitted for brevity. Finally, $P_i$ is projected via a classifier layer to generate the continuous change probability map $M_i$.

### E. Optimization Objective

The proposed framework is optimized end-to-end via an adversarial minimax game between the generator $G$ and the domain discriminator $D$. A composite objective is designed to synergistically train the network for both robust cross-modal feature alignment and high-precision CD.

The generator is optimized via a composite loss $\mathcal{L}_G$. Serving as the core component, the CD loss $\mathcal{L}_{\mathrm{CD}}$ implements deep supervision across hierarchical continuous predictions $\{M_i\}_{i=1}^4$ to guarantee effective gradient propagation

$$\mathcal{L}_{\mathrm{CD}} = \sum_{i=1}^{4} \mathcal{L}_{\mathrm{base}}\left(M_i, \mathrm{Down}_i\left(Y\right)\right) \tag{18}$$

where $Y$ is the ground truth, $\mathrm{Down}_i(\cdot)$ represents resolution downsampling, and $\mathcal{L}_{\mathrm{base}}$ combines BCE and Dice loss to counteract class imbalance.

In addition to pixel-level supervision, the generator is constrained by an adversarial loss $\mathcal{L}_{\mathrm{adv}}$ to enforce modality invariance and a geometric contrastive loss $\mathcal{L}_{\mathrm{PAFR}}$ for manifold structure regularization. This synergistic formulation ensures the network learns features that are both aligned across modalities and discriminative for the CD task. Consequently, the total generator objective is formulated as

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{CD}} + \lambda_{\mathrm{adv}} \sum_{i=1}^{3} \mathcal{L}_{\mathrm{adv}}^i + \lambda_{\mathrm{con}} \sum_{i=1}^{3} \mathcal{L}_{\mathrm{PAFR}}^i \tag{19}$$

where $\lambda_{\mathrm{adv}} = 0.001$ and $\lambda_{\mathrm{con}} = 0.002$ balance the contributions of the adversarial and geometric regularization losses, respectively. Because both losses are aggregated over multiple dense feature pyramids, their raw gradients are exceptionally large, and their coefficients must be strictly scaled down to prevent these auxiliary gradients from overwhelming the primary pixel-level segmentation objective. Furthermore, a slightly stronger regularization push is required to preserve fine-grained structural boundaries, ensuring they function purely as gentle manifold regularizers, guiding the latent space distribution without disrupting the stable convergence of the primary CD task. A comprehensive ablation study empirically verifying this gradient balancing strategy is provided in Section V-E.

Concurrently, the multiscale discriminator $D = \{D_i\}_{i=1}^3$ is trained to distinguish between source domain $(\mathcal{S})$ and target domain $(\mathcal{T})$ conditioned feature distributions. To ensure training stability, we employ the hinge adversarial loss augmented with an $R_1$ gradient penalty. The total discriminator objective $\mathcal{L}_D$ is aggregated across the three alignment scales

$$\begin{aligned}\mathcal{L}_D = \sum_{i=1}^{3} \Big( & \mathbb{E}_{H_{\mathcal{S}}^i}\left[\max\left(0, 1 - D_i\left(H_{\mathcal{S}}^i\right)\right)\right] \\ & + \mathbb{E}_{H_{\mathcal{T}}^i}\left[\max\left(0, 1 + D_i\left(H_{\mathcal{T}}^i\right)\right)\right] \\ & + \mathbb{E}_{H_{\mathcal{S}}^i}\left[\left\|\nabla D_i\left(H_{\mathcal{S}}^i\right)\right\|_2^2\right]\Big)\end{aligned} \tag{20}$$

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where $H_{\mathcal{S}}^{i}$ and $H_{\mathcal{T}}^{i}$ denote conditioned joint feature maps from source domain $\mathcal{S}$ and target domain $\mathcal{T}$ at scale $i$, respectively. The interplay between minimizing $\mathcal{L}_G$ and maximizing $\mathcal{L}_D$ drives the end-to-end optimization of the network.

## V. Experiments and Analysis

In this section, comprehensive experiments are conducted to assess the effectiveness and robustness of ASFR-Net. We first detail the experimental setup, including benchmark datasets, implementation details, and evaluation metrics. Subsequently, quantitative and qualitative comparisons are conducted against a suite of representative and advanced methods. Finally, we perform ablation studies to validate the impact of core components, backbone selections, and key hyperparameters.

### A. Datasets

The generalization of our method is evaluated on three representative benchmarks. Spanning visible-NIR and optical-SAR modalities, these datasets possess distinct modal disparities, providing diverse and challenging evaluation scenarios.

*1) VisNIR-HCD Dataset (Ours):* As detailed in Section III, this is a high-resolution benchmark comprising 8432 visible-NIR image pairs focused on building changes. It presents a significant challenge in handling nonlinear spectral heterogeneity, particularly the distinct reflectance of vegetation and man-made structures between the RGB and NIR bands. For our experiments, we strictly adhere to the official partition of 5901 training, 839 validation, and 1692 test pairs.

*2) MT-Wuhan Dataset [18]:* This widely-used optical-SAR benchmark captures land-cover changes resulting from rapid urbanization in Wuhan, China. The dataset is partitioned into 552 training, 129 validation, and 112 test pairs of 256 $\times$ 256 pixels, where the number of unchanged pixels is approximately five times that of the changed pixels. Despite the limited sample size, its complex urban textures present a rigorous test for model generalization under data scarcity conditions.

*3) XiongAn Dataset [12]:* This dataset focuses on building CD during large-scale urban redevelopment in Xiong'an New Area, China, comprising GaoFen-2 multispectral and GaoFen-3 SAR images. The inherent speckle noise and geometric distortions of SAR imagery pose severe challenges to detection accuracy. As the official test set is not publicly available, we follow common practice and use the validation set for evaluation. The original images have a resolution of 512 $\times$ 512 pixels. In our experiments, we crop them into nonoverlapping 256 $\times$ 256 patches, ultimately resulting in 7604 pairs for training, and 1652 pairs for validation and testing, respectively.

### B. Implementation Details and Evaluation Metrics

Our approach is implemented in PyTorch, with all experiments conducted on a workstation equipped with two NVIDIA GeForce RTX 3090 graphics processing units (GPUs) (24 GB). We adopt a lightweight MobileNetV2 [23] backbone pretrained on ImageNet as the weight-sharing feature extractor, while other components are randomly initialized. The network is optimized using the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$, and a weight decay of $1e-2$. The initial learning rate is set to 0.0005 with a batch size of 16. A Poly learning rate schedule with a power of 0.9 is employed throughout training. The model is trained for 40, 2, and 20k iterations on the VisNIR-HCD, MT-Wuhan, and XiongAn datasets, respectively. A variety of online data augmentation techniques, including random rescaling, cropping, flipping, temporal exchange, and cutmix are incorporated to enhance generalization. The model is evaluated on the validation set after each epoch, saving the checkpoint with the highest $F1$-score ($F1$) for final testing.

In line with standard practices, five evaluation metrics are employed to quantify performance: precision (Pre), recall (Rec), $F1$, intersection over union (IoU), and overall accuracy (OA). Among these, $F1$ and IoU are considered the primary indicators for comprehensive evaluation.

### C. Quantitative Analysis

Table II details the quantitative comparison results of ASFR-Net against related methods, including A2Net [27], RFANet [25], STADE-CDNet [28], MambaCD [29], STENet [30], DGMA2-Net [31], CSI-Net [32], AFENet [11], CASP [33], ConvFormer-CD [34], SFEARNet [35], EFICNN [36], HeteCD [12], and HRMNet [37], on the three datasets. As evidently observed, ASFR-Net consistently achieves superior performance across key metrics, demonstrating its exceptional robustness in handling diverse heterogeneous scenarios.

In the challenging VisNIR-HCD dataset, where nonlinear spectral discrepancies between the visible and NIR bands often induce significant pseudochanges, ASFR-Net secures a leading $F1$ of 80.44% and an IoU of 67.28%. This performance substantially surpasses traditional CNN-based Siamese networks such as RFANet [25] and A2Net [27]. It is worth noting that while the registration-aware method CASP [33] performs competitively on other datasets, its performance drops on VisNIR-HCD with an $F1$ of 76.43%. This indicates that relying solely on spatial alignment is insufficient to bridge the profound visible-NIR spectral gap. Furthermore, compared to the heterogeneous-specific method AFENet [11], which yields an $F1$ of 68.51%, our approach achieves a remarkable improvement of nearly 12%. This significant margin suggests that the global adversarial strategy employed by AFENet tends to oversmooth high-frequency details, whereas our coarse-to-fine strategy effectively preserves critical boundary information while mitigating spectral inconsistencies through the frequency-domain refinement module.

Extending the evaluation to optical-SAR scenarios, ASFR-Net exhibits exceptional resistance to inherent speckle noise and geometric distortions found in the MT-Wuhan and XiongAn datasets. On the noise-heavy MT-Wuhan benchmark, our method outperforms top competitors like EFICNN [36] and DGMA2-Net [31] by approximately 2% in terms of $F1$. Furthermore, it is worth noting that although lightweight models such as A2Net [27] establish a competitive baseline on VisNIR-HCD by exploiting spatial structural similarity, relying solely on spatial alignment is a fragile strategy for heterogeneous tasks. This limitation remains largely concealed in the XiongAn dataset due to its clean multipolarization SAR

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TABLE II

QUANTITATIVE RESULTS OF DIFFERENT COMPARISON METHODS ON THE VISNIR-HCD, MT-WUHAN, AND XIONGAN DATASETS. THE EVALUATION METRICS INCLUDE MODEL COMPLEXITY [PARAMS (M) AND FLOPS (G)] AND DETECTION ACCURACY [PRECISION (PRE), RECALL (REC), $F1$, IOU, AND OA]. COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

| Methods | Params (M)↓ | FLOPs (G)↓ | VisNIR-HCD (%) | | | | | MT-Wuhan (%) | | | | | XiongAn (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ |
| A2Net [27] | 3.78 | 3.05 | **82.54** | 75.22 | **78.71** | **64.89** | **98.77** | 56.89 | 52.81 | 54.78 | 37.72 | 87.80 | 84.82 | 82.19 | **83.49** | **71.65** | 98.99 |
| RFANet [25] | 2.86 | 3.16 | 82.17 | **75.30** | 78.58 | 64.72 | **98.76** | 56.74 | 58.82 | 57.76 | 40.61 | 87.97 | 82.06 | **83.83** | 82.94 | 70.85 | 98.93 |
| STADE-CDNet [28] | 3.50 | 11.99 | 74.84 | 51.49 | 61.01 | 43.90 | 98.02 | 32.62 | 53.80 | 40.62 | 25.49 | 78.00 | 56.57 | 53.80 | 55.15 | 38.07 | 97.28 |
| MambaCD [29] | 85.53 | 44.83 | 76.43 | 74.24 | 75.32 | 60.41 | 98.53 | 58.50 | 54.63 | 56.50 | 39.37 | 88.24 | 79.20 | **86.20** | 82.55 | 70.29 | 98.87 |
| STENet [30] | 10.95 | 14.73 | 80.53 | 68.67 | 74.13 | 58.89 | 98.56 | 42.93 | 51.66 | 46.89 | 30.63 | 83.64 | 81.66 | 80.01 | 80.83 | 67.83 | 98.82 |
| DGMA2-Net [31] | 37.10 | 18.10 | 80.81 | 73.39 | 76.92 | 62.50 | 98.67 | 52.27 | **66.97** | **58.72** | **41.56** | 86.83 | 83.05 | 82.30 | 82.67 | 70.46 | 98.93 |
| CSI-Net [32] | 62.18 | 367.25 | 75.63 | 59.18 | 66.40 | 49.70 | 98.20 | 43.83 | 49.95 | 46.69 | 30.46 | 84.04 | 79.53 | 69.12 | 73.96 | 58.68 | 98.57 |
| AFENet [11] | 39.7 | 398.58 | 68.23 | 68.79 | 68.51 | 52.11 | 98.09 | 41.79 | **69.26** | 52.13 | 35.25 | 82.21 | 81.18 | 80.92 | 81.05 | 68.14 | 98.82 |
| CASP [33] | 1.74 | 2.63 | 81.28 | 72.16 | 76.43 | 61.86 | 98.66 | **63.48** | 47.90 | 54.60 | 37.55 | **88.86** | **86.78** | 80.30 | 83.41 | 71.55 | **99.01** |
| ConvFormer-CD [34] | 37.72 | 5.14 | 69.44 | 67.38 | 68.40 | 51.97 | 98.10 | **62.43** | 45.77 | 52.81 | 35.88 | 88.52 | 84.03 | 79.46 | 81.68 | 69.03 | 98.89 |
| SFEARNet [35] | 5.56 | 4.65 | 78.89 | 75.09 | 76.94 | 62.53 | 98.64 | 54.61 | 61.22 | 57.73 | 40.57 | 87.46 | **85.76** | 80.39 | 82.99 | 70.93 | 98.97 |
| EFICNN [36] | 21.32 | 91.34 | 81.93 | **75.87** | **78.78** | **64.99** | **98.77** | 60.16 | 57.74 | **58.92** | **41.77** | **88.74** | 85.43 | 81.14 | 83.23 | 71.28 | 98.98 |
| HeteCD [12] | 63.85 | 81.28 | 75.61 | 74.05 | 74.82 | 59.77 | 98.50 | **61.58** | 55.45 | 58.35 | 41.19 | **88.93** | 85.05 | 80.32 | 82.62 | 70.38 | 98.95 |
| HRMNet [37] | 13.46 | 12.17 | **82.41** | 74.08 | 78.02 | 63.96 | 98.74 | 52.65 | 55.07 | 53.83 | 36.83 | 86.79 | 84.74 | **82.80** | **83.76** | **72.06** | **99.00** |
| Ours | 6.35 | 15.13† | **82.45** | **78.53** | **80.44** | **67.28** | **98.86** | 58.68 | **62.99** | **60.76** | **43.64** | 88.62 | **87.17** | 82.66 | **84.86** | **73.71** | **99.08** |

†Note: Since standard profiling tools often fail to track hardware-level FLOPs for frequency-domain operations, for methods incorporating ASAU, we manually calculated this overhead based on the Cooley-Tukey algorithm: $\mathcal{O}(C \cdot HW \log_2(HW))$. By meticulously accumulating this across all 5 pyramid levels in our Siamese architecture, the FFT/iFFT operations contribute approximately 0.39 GFLOPs, which has been added to the FLOPs of ASFR-Net to ensure transparency of complexity reporting.

imagery and relatively precise annotations. However, it is fully exposed during evaluation on the MT-Wuhan dataset, yielding an $F1$ score of only 54.78% for A2Net. In stark contrast, ASFR-Net demonstrates modality-agnostic robustness across both RGB-NIR and RGB-SAR scenarios. By treating modal disparities as generalized domain shifts in the latent space and adaptively decoupling task-irrelevant artifacts in the Fourier domain, ASFR-Net effectively overcomes the fragility of pure spatial alignment without relying on modality-specific priors.

Beyond raw detection accuracy, computational efficiency is a critical factor for practical deployment in real-world RS applications. As detailed in Table II, many existing high-performance models come with prohibitive computational costs. For instance, MambaCD [29] and AFENet [11] require massive parameters exceeding 85 and 39 M, respectively, along with high FLOPs, which severely hinders their deployability on edge devices. Conversely, ASFR-Net adopts a relatively efficient design with only 6.35 M parameters and 15.13 G FLOPs. Although our model incurs a marginal increase in computational overhead compared to existing ultra-lightweight networks like RFANet [25], this tradeoff is highly justified by the significant performance gains. In particular, ASFR-Net outperforms RFANet by a solid margin of 2.56% in terms of IoU on the VisNIR-HCD dataset. Consequently, ASFR-Net establishes an optimal equilibrium between model complexity and detection accuracy, ensuring high-precision inference while maintaining a manageable memory footprint suitable for large-scale data processing.

### D. Qualitative Analysis

To visually assess and compare the detection performance among different CD approaches, we also carry out qualitative visualizations on the three datasets, which are presented in Fig. 7. The qualitative results presented in these figures illustrate that our method exhibits superiority over other CD methods.

In the VisNIR-HCD dataset, significant spectral reflectance differences in vegetation and building roofs between RGB and NIR bands constitute the primary source of pseudochanges. Empirical observations reveal that methods such as AFENet [11], STADE-CDNet [28], and ConvFormer-CD [34] fail to effectively decouple these spectral discrepancies, resulting in dense false positives (FPs) in vegetation-covered areas. Similarly, MambaCD [29] frequently misclassifies seasonal vegetation variations as building changes. On the contrary, the prediction maps generated by ASFR-Net exhibit the highest purity, characterized by coherent change regions and minimal FP noise. This superior robustness demonstrates that GADU successfully aligns heterogeneous features via modality-invariant learning, while SFEM leverages frequency-domain filtering to further suppress residual spectral deviations, thereby enabling the model to focus on genuine semantic structural changes.

For the MT-Wuhan and XiongAn datasets, the inherent SAR speckle noise often compromises model inference. STADE-CDNet [28] and STENet [30] falter significantly when processing these samples, misidentifying the granular texture of SAR as changes. In the complex scenes of XiongAn, particularly in rows 10 and 11, these methods generate extensive blocky false alarms marked in red. Moreover, DGMA2-Net [31] and CASP [33] exhibit marked discontinuities and missed detections (FNs) when detecting the slender road in row 8, revealing the limitations of context modeling under strong noise. Conversely, ASFR-Net demonstrates excellent noise resistance and object completeness. As shown in rows 7 and 11 of the figure, it successfully suppresses background speckles that confuse other models and accurately extracts complete change shapes. This performance is attributable to

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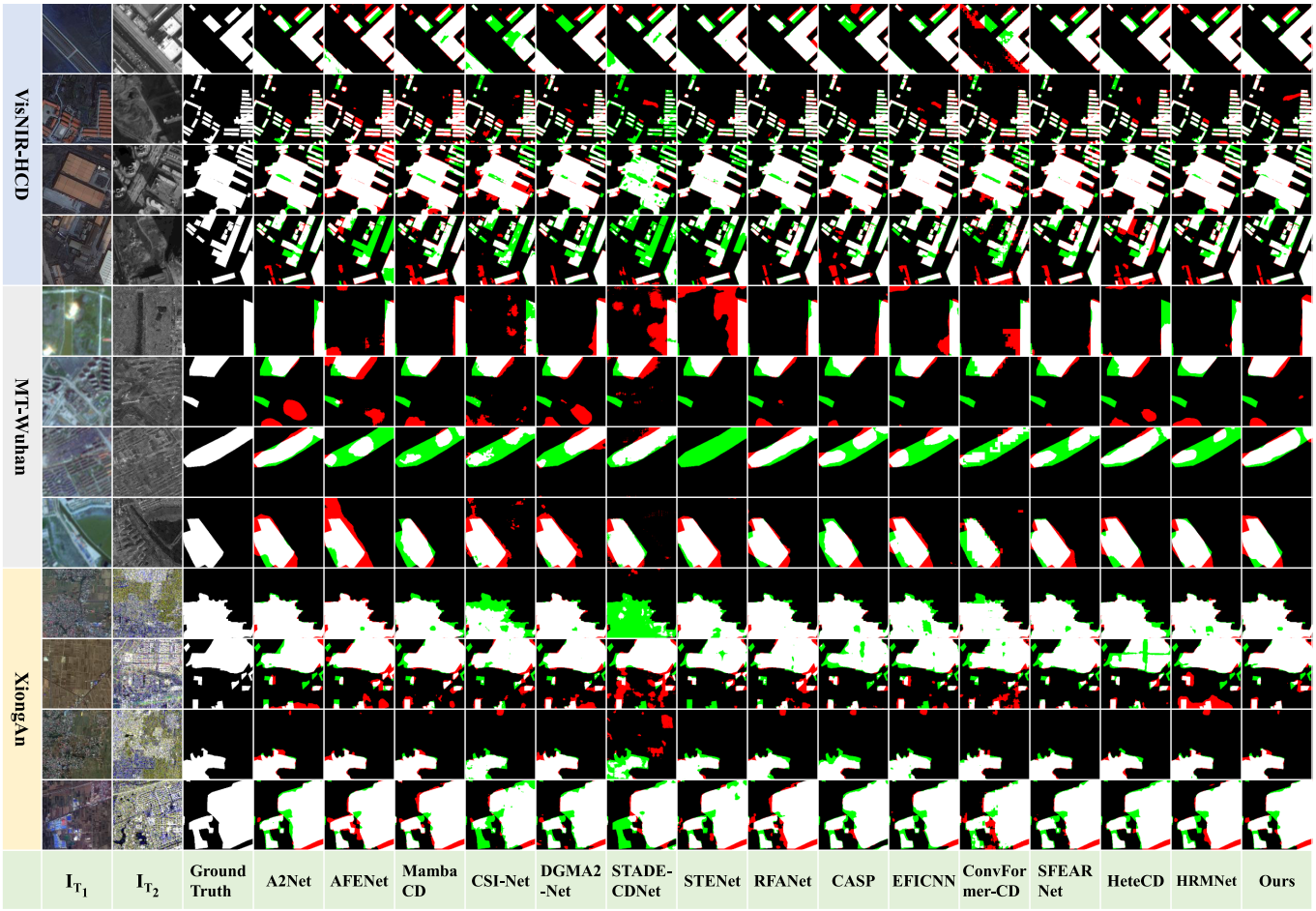



Fig. 7. Qualitative visualizations of different methods tested on VisNIR-HCD, MT-Wuhan, and XiongAn datasets. The visualizations employ a standard color scheme for error analysis: true positives (TPs) in white, true negatives (TNs) in black, FPs in red, and FNs in green.

the frequency-domain purification and multiscale aggregation strategies in SFEM, ensuring that the generated detection map is internally coherent and contiguous, significantly reducing the false alarm rate while ensuring a competitive recall.

In urban monitoring tasks, accurately delineating building boundaries and separating closely adjacent targets are critical for evaluating model practicality. In the dense urban scenes of the XiongAn dataset (row 12), affected by the geometric distortion of SAR side-looking imaging, the prediction maps generated by RFANet [25] and EFICNN [36] suffer from boundary erosion and incomplete detection, represented by false negatives (FNs). HRMNet [37] also manifests edge blurring when detecting large factories in VisNIR-HCD. On the contrary, ASFR-Net exhibits superior boundary preservation in all scenarios. Even in the high-density building area of XiongAn, it generates sharp and regular contours and clearly separates adjacent targets. This precision is primarily attributed to the synergy between the deep semantic guidance of HGFM and the explicit geometric constraints of the PAFR loss. By regulating class margins in the feature space, ASFR-Net effectively suppresses prevalent blurring artifacts and ensures high structural fidelity.

Despite the superior qualitative and quantitative performance, ASFR-Net is constrained by a fundamental limitation inherent to cross-modal tasks: it cannot resolve changes that are inherently unobservable in one modality. The practical boundary for ASFR-Net to work reliably relies on the premise that a genuine semantic change must leave an observable, structure-related imprint, such as clear edges or regular geometries, in both modalities, even if their radiometric relationship is nonlinear. Consequently, there are inevitable modal blind spots where the method may experience performance degradation, as evidenced by the minor error regions in Fig. 7. The first failure scenario is low-contrast structure invisibility, leading to missed detections (FNs). For instance, in row 3 of VisNIR-HCD, a distinct green gap appears at the edge of the large building on the right side of our prediction map. Observing the corresponding $I_{T_2}$ reveals that this roof section exhibits lower reflectance, making it visually indistinguishable from the adjacent dark ground or shadows. The second scenario is deceptive structure interference, causing false alarms (FPs). This occurs when large-scale nonbuilding events generate strong, regular geometric patterns in a specific modality, coincidentally mimicking the structural characteristics of buildings. As observed in row 10 of Fig. 7, several distinct red patches appear below the correctly detected white region. Although $I_{T_1}$ shows bare land in that area, $I_{T_2}$ exhibits highly structured, bright scattering signals. These strong nonbuilding geometric

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TABLE III

ABLATION STUDY FOR DIFFERENT COMPONENTS ON THE THREE DATASETS. COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

| Methods | Params (M)↓ | FLOPs (G)↓ | VisNIR-HCD (%) | | | | | MT-Wuhan (%) | | | | | XiongAn (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ |
| (a) Baseline | 2.63 | 2.54 | 81.75 | 73.84 | 77.59 | 63.39 | 98.72 | 53.01 | 58.23 | 55.50 | 38.41 | 86.94 | 82.92 | 84.35 | 83.63 | 71.87 | 98.97 |
| (b) + MIR-Learner | 4.50 | 3.65 | 85.16 | 74.35 | 79.38 | 65.82 | 98.84 | 62.29 | 52.94 | 57.23 | 40.09 | 88.94 | 85.13 | 82.69 | 83.89 | 72.25 | 99.01 |
| (c) + MIR-Learner (w/o PAFR) | 4.50 | 3.65 | 81.42 | 76.59 | 78.93 | 65.20 | 98.77 | 57.73 | 55.91 | 56.80 | 39.67 | 88.11 | 87.12 | 80.58 | 83.72 | 72.00 | 99.03 |
| (d) + SFEM | 3.77 | 10.92 | 82.47 | 76.46 | 79.35 | 65.77 | 98.80 | 52.67 | 63.26 | 57.48 | 40.33 | 86.91 | 83.91 | 84.75 | 84.33 | 72.91 | 99.02 |
| (e) + SFEM (w/o HPAU) | 2.68 | 3.11 | 80.00 | 76.41 | 78.16 | 64.16 | 98.71 | 55.70 | 58.86 | 57.23 | 40.09 | 87.70 | 82.81 | 85.67 | 84.21 | 72.73 | 99.00 |
| (f) + SFEM (w/o ASAU) | 3.49 | 8.49 | 81.86 | 76.62 | 79.15 | 65.50 | 98.78 | 57.42 | 57.07 | 57.24 | 40.10 | 88.08 | 85.97 | 82.32 | 84.11 | 72.57 | 99.03 |
| (g) + HGFM | 2.69 | 2.85 | 81.17 | 77.91 | 79.50 | 65.98 | 98.84 | 53.97 | 59.49 | 56.60 | 39.47 | 87.24 | 87.12 | 81.26 | 84.09 | 72.54 | 99.04 |
| (h) + MIR-Learner + SFEM | 5.65 | 12.03 | 85.91 | 74.37 | 79.72 | 66.28 | 98.86 | 57.11 | 58.04 | 58.08 | 40.92 | 88.28 | 85.22 | 83.94 | 84.57 | 73.27 | 99.05 |
| (i) + MIR-Learner + HGFM | 4.56 | 3.96 | 84.60 | 75.88 | 80.00 | 66.67 | 98.86 | 60.92 | 57.10 | 58.95 | 41.80 | 88.88 | 85.55 | 83.37 | 84.45 | 73.08 | 99.04 |
| (j) + SFEM + HGFM | 3.83 | 11.23 | 82.32 | 78.27 | 80.24 | 67.01 | 98.84 | 62.21 | 56.77 | 59.37 | 42.21 | 89.13 | 85.56 | 83.55 | 84.55 | 73.23 | 99.05 |
| (k) ASFR-Net (Ours) | 6.35 | 15.13 | 82.45 | 78.53 | 80.44 | 67.28 | 98.86 | 58.68 | 62.99 | 60.76 | 43.64 | 88.62 | 87.17 | 82.66 | 84.86 | 73.71 | 99.08 |

TABLE IV

COMPARISON OF DIFFERENT BACKBONE ARCHITECTURES ON THE THREE DATASETS. COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

| Methods | Params (M)↓ | FLOPs (G)↓ | VisNIR-HCD (%) | | | | | MT-Wuhan (%) | | | | | XiongAn (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ |
| (l) ResNet18 [38] | 16.35 | 21.09 | 81.14 | 78.71 | 79.90 | 66.53 | 98.81 | 56.59 | 57.86 | 57.22 | 40.07 | 87.90 | 84.77 | 84.20 | 84.48 | 73.14 | 99.04 |
| (m) EfficientNet [39] | 8.18 | 18.41 | 83.06 | 77.43 | 80.15 | 66.87 | 98.84 | 61.45 | 55.41 | 58.27 | 41.12 | 88.90 | 86.92 | 82.01 | 84.39 | 73.00 | 99.06 |
| (n) MobileViT [40] | 9.80 | 18.90 | 79.97 | 70.33 | 74.84 | 59.80 | 98.58 | 49.97 | 54.67 | 52.21 | 35.33 | 86.01 | 83.66 | 82.63 | 83.14 | 71.15 | 98.96 |
| (o) ConvNeXt [41] | 34.56 | 64.48 | 83.30 | 76.17 | 79.58 | 66.08 | 98.82 | 56.84 | 54.27 | 55.52 | 38.43 | 87.84 | 86.54 | 82.65 | 84.55 | 73.24 | 99.06 |
| (p) LWGANet [42] | 23.67 | 21.22 | 84.77 | 76.46 | 80.40 | 67.23 | 98.88 | 62.73 | 56.22 | 59.29 | 42.14 | 89.21 | 86.99 | 83.03 | 84.96 | 73.85 | 99.09 |

patterns deceive the network. Ultimately, ASFR-Net cannot intrinsically resolve fundamentally unobservable changes; rather, it suppresses residual modal disparities, provided that the underlying structural evidence of the targets remains intact.

### E. Ablation Study

To verify the effectiveness and necessity of the core components in ASFR-Net, we conduct detailed ablation experiments on all three datasets. The evaluated components include MIR-Learner, SFEM, and HGFM. Furthermore, we also evaluate the performance of ASFR-Net when incorporating different backbone alternatives as substitutions. The quantitative results are summarized in Tables III and IV. The baseline [model (a)] is defined as a weight-sharing Siamese MobileNetV2 backbone with basic difference fusion and a simplified feature pyramid network decoder, without the proposed modules.

*1) Effectiveness of Core Components:* We first evaluate the effectiveness of MIR-Learner. Incorporating MIR-Learner in model (b) substantially improves the $F1$ from 77.59% to 79.38% on the VisNIR-HCD dataset. To isolate the specific contribution of the geometric constraint, model (c) is evaluated by removing the PAFR. This exclusion leads to a notable decline in pre on VisNIR-HCD from 85.16% to 81.42%. This shows that while the adversarial component aligns global distributions, the PAFR is indispensable for maintaining feature discriminability by enforcing a margin between changed and unchanged features in the latent space. The t-SNE visualization in Fig. 8 further corroborates that MIR-Learner effectively aligns distributions while preserving semantic separability.

Building upon the aligned features, SFEM is introduced to suppress residual modal noise and recover fine-grained structural details. The integration of the full SFEM in model (d) yields significant quantitative gains, particularly in Rec, which increases from 58.23% to 63.26% on the MT-Wuhan dataset. Removing HPAU in model (e) results in a loss of structural integrity, and the exclusion of ASAU in model (f) degrades performance, suggesting that the network struggles to filter sensor-specific noise without frequency-domain analysis. These quantitative comparisons validate that ASAU and HPAU work synergistically to decouple modal artifacts from genuine changes. To intuitively comprehend this refinement process, Fig. 9 visualizes the multiscale feature heatmaps for three examples. As clearly observed, the features extracted by the baseline model ($B_t^i$) suffer from severe modal interference, heavily activating on background noise such as SAR speckles

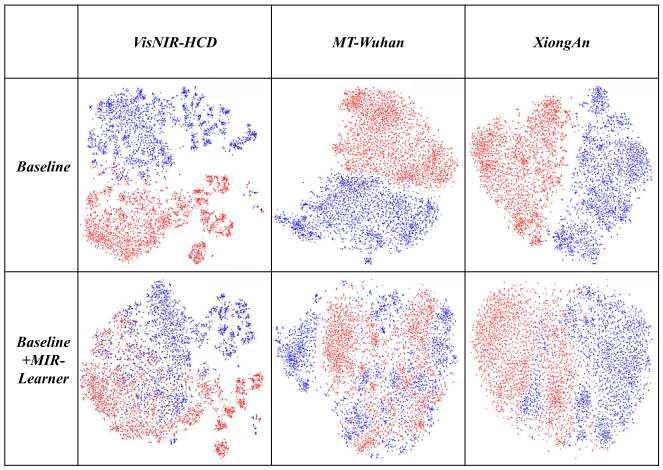



Fig. 8. Comparative t-SNE visualization of feature distributions with and without the MIR-Learner. The blue and red points correspond to features from two heterogeneous modalities, such as optical and SAR or NIR data.

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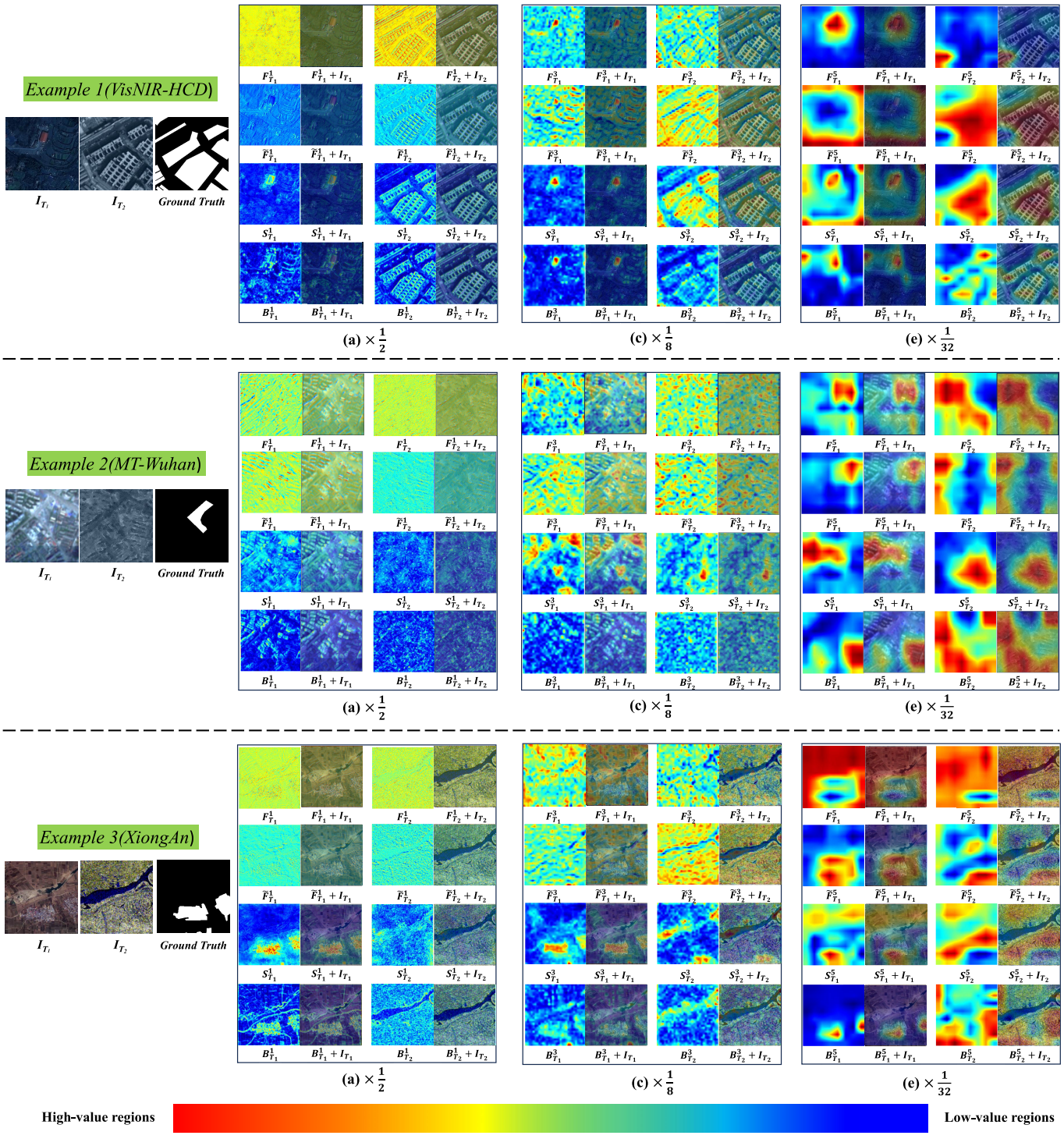



Fig. 9. Progressive heatmap visualizations of multiple levels for ASFR-Net on the three datasets. $I_{T_1}$ and $I_{T_2}$ are the bi-temporal images, followed by the corresponding ground truth. (a), (c), and (e) Visualize the features from three distinct hierarchical levels, corresponding to downsampling rates of 1/2, 1/8, and 1/32, respectively. Each panel sequentially displays four feature states from top to bottom: the raw features $F_t^i$ extracted by the backbone, the intermediate features $\tilde{F}_t^i$ purified by ASAU, the reinforced features $S_t^i$ aggregated by HPAU, and the features $B_t^i$ generated by the baseline [model (a)] in Table III, which utilizes basic convolutions for alignment. Each state is presented alongside its composite visualization overlaid on the original image, such as $F_t^i + I_t$. A standard jet color map is employed, where warm colors indicate high activation values and cool colors represent low activation or suppressed regions.

or spectral deviations. On the contrary, within our framework, the raw backbone features ($F_t^i$) first capture the coarse regions. After being explicitly purified by the ASAU in the frequency domain ($\tilde{F}_t^i$) and structurally aggregated by the HPAU ($S_t^i$), the task-irrelevant modal artifacts are effectively suppressed. Consequently, the final representations ($S_t^i$) strictly focus on genuine semantic structural changes, visually corroborating the superiority of the proposed SFEM.

Furthermore, replacing the baseline decoder with the proposed HGFM blocks in model (g) provides consistent

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TABLE V
HYPERPARAMETER SENSITIVITY ANALYSIS ON THE VISNIR-HCD DATASET. COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

| Settings | $\alpha$ | $\beta$ | $\lambda_{adv}$ | $\lambda_{con}$ | Pre ↑ | Rec ↑ | F1 ↑ | IoU ↑ | OA ↑ |
|---|---|---|---|---|---|---|---|---|---|
| $\alpha$ too small | 0.5 | 0.05 | 0.001 | 0.002 | **84.46** | 76.68 | **80.38** | **67.20** | **98.87** |
| $\alpha$ too strict | 1.5 | 0.05 | 0.001 | 0.002 | **84.23** | 76.46 | 80.15 | 66.88 | **98.86** |
| Zero tolerance | 1.0 | 0.0 | 0.001 | 0.002 | 81.10 | **79.25** | 80.16 | 66.90 | 98.82 |
| $\beta$ too loose | 1.0 | 0.2 | 0.001 | 0.002 | 81.90 | 78.12 | 79.96 | 66.62 | 98.82 |
| Wrong magnitude | 1.0 | 0.05 | 0.01 | 0.02 | 83.60 | 76.72 | 80.01 | 66.68 | 98.84 |
| Ratio inverted | 1.0 | 0.05 | 0.002 | 0.001 | 84.07 | 76.19 | 79.93 | 66.57 | 98.85 |
| **Ours** | **1.0** | **0.05** | **0.001** | **0.002** | 82.45 | **78.53** | **80.44** | **67.28** | **98.86** |

improvements across all metrics. For instance, on the XiongAn dataset, HGFM improves the IoU from 71.87% to 72.54%, demonstrating that the top-down guidance strategy effectively bridges the semantic-spatial gap by using deep semantics to refine the boundaries of shallow difference features.

The complete ASFR-Net (model k), which integrates all the aforementioned modules, achieves the best overall performance across all benchmarks. The synergistic effect is clearly demonstrated when observing the performance degradation in combinations where key components are absent. For example, combining only SFEM and HGFM in model (j) without MIR-Learner leads to a sharp drop in $F1$ on MT-Wuhan from 60.76% to 59.37%, highlighting the necessity of adversarial alignment for bridging large modal gaps.

*2) Choice of Backbone Architectures:* Finally, Table IV assesses different backbone architectures to justify our design choice. Compared to heavier CNNs, namely ResNet18 [38] and ConvNeXt [41], our MobileNetV2-based model achieves superior accuracy with significantly fewer parameters. Training logs further reveal that overparameterized models suffer from validation fluctuations, likely overfitting to sensor-specific modal noise rather than learning generalizable cross-modal semantics. Among efficient alternatives, EfficientNet [39] delivers a highly competitive $F1$ score but incurs higher computational overhead. The recently proposed RS backbone LWGANet [42] demonstrates remarkable capabilities. Although its parameter count is notably larger than that of MobileNetV2, its efficient grouped attention mechanism ensures that the actual GPU memory footprint as well as the training and inference times remain surprisingly comparable to our lightweight design. Furthermore, the final performance metrics of LWGANet are highly competitive and almost identical to our baseline model across most evaluation tracks. Nevertheless, our MobileNetV2-based architecture maintains a slight advantage on the highly noisy MT-Wuhan dataset while requiring nearly a quarter of the parameters. Consequently, considering the experimental outcomes across the three datasets, MobileNetV2 is identified as the optimal architecture for this task, as it achieves the best overall detection accuracy and superior training stability while simultaneously demanding the lowest computational overhead.

*3) Hyperparameter Sensitivity Analysis:* To justify the selected hyperparameter configurations, we evaluated ASFR-Net under specific boundary variations on the VisNIR-HCD dataset, as summarized in Table V.

First, regarding the geometric boundaries ($\alpha$ and $\beta$) in the PAFR module: In the $L_2$-normalized feature space, setting $\alpha = 1.0$ guarantees an angular separation of at least 60° between changed feature pairs. As shown in Table V, reducing $\alpha$ to 0.5 fails to push changed features far enough apart, causing a slight drop in $F1$. Conversely, setting $\alpha = 1.5$ imposes an overly strict constraint, disrupting the feature manifold optimization and degrading $F1$–80.15%. For the attraction margin $\beta$, setting it to zero tolerance of 0.0 will force the network to penalize inherent sensor noise, causing modality-specific overfitting and a noticeable Pre drop. Setting $\beta = 0.2$ results in loose clustering, generating excessive FPs and dropping the overall $F1$ to 79.96%.

Second, regarding the gradient balancing coefficients ($\lambda_{adv}, \lambda_{con}$): these parameters dictate the multitask optimization dynamics. When scaled up to the wrong magnitude, the massive gradients generated by the dense multiscale adversarial and contrastive pyramids disrupt the pixel-level segmentation loss, leading to suboptimal convergence, with $F1$ dropping to 80.01%. When the ratio is inverted ($\lambda_{adv} = 0.002 > \lambda_{con} = 0.001$), the aggressive global distribution alignment overshadows the local geometric constraints. This overalignment inadvertently smooths out fine-grained boundary details, degrading $F1$ to 79.93%. Thus, maintaining the $10^{-3}$ magnitude and setting $\lambda_{con} > \lambda_{adv}$ provides a slightly stronger contrastive push, acting as the optimal gentle manifold regularizer.

## VI. CONCLUSION AND FUTURE WORK

In this article, we propose ASFR-Net, a robust framework designed to reconcile the intrinsic conflict between feature alignment and semantic discriminability in heterogeneous CD. Adopting a coarse-to-fine refinement paradigm, the model synergizes adversarial learning with frequency-domain analysis. In particular, MIR-Learner mitigates the primary modal discrepancy by enforcing geometric regularity, while SFEM demonstrates that residual modal discrepancies, which are often indistinguishable in the spatial domain, can be effectively decoupled and suppressed via frequency-domain priors. Then, HGFM bridges the semantic-spatial gap by employing hierarchical guided fusion, ensuring that deep semantic priors effectively sharpen fine-grained details for precise boundary delineation. Furthermore, the VisNIR-HCD benchmark alleviates the data scarcity issue in multisource building CD. Extensive empirical evaluations substantiate that ASFR-Net achieves SOTA performance, demonstrating superior robustness against complex spectral heterogeneity and noise.

Although the current framework achieves remarkable performance, its practical application in complex real-world scenarios remains constrained by strict spatial image alignment. Future work will be dedicated to breaking this limitation, further enhancing the robustness and practicality of the model. Since perfectly aligned heterogeneous image pairs are scarce in rapid-response scenarios due to sensor parallax, jitter, and imaging principle differences, we plan to explore a unified architecture that jointly optimizes cross-modal registration and CD in a mutually reinforcing manner. By investigating advanced representation disentanglement strategies, we aim to

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effectively separate the features required for spatial alignment from those critical for change identification. Meanwhile, we will explore task-driven feedback mechanisms, allowing the downstream CD process to actively guide and refine the spatial registration. Through this joint paradigm, we expect to minimize misalignment-induced pseudo-changes, mitigate the impact of adverse artifacts, and ultimately prevent the performance degradation of downstream semantic tasks.

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